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GENESIS

Genesis Ideathon: Final Presentation

Team ID: *[GEN26-A0H-I213]*

Team Name: *[THE ALCHESMISTS]*

Domain: *[Space, Aerospace, Mobility, and Next Gen Infrastructure]*

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Problem Statement

[Problem statement: Today bridges are inspected manually from time to time and assumed to be safe in between. This yields to three issues:
The state of bridges in between manual inspections cannot be assessed.
Even if a problem is reported, there is no force majeure to restrict their use.
No one is accountable and documentation can be tampered with. Reports of inspections or the marking of issues as "resolved" can be altered, and after an incident it may prove difficult to establish who knew what and had knowledge of a potential danger, and who approved the continued use of the bridge.
As a result, unsafe bridges continue to be used until they are proven to be safe structures.]

Problem Understanding

- [Commuters, pedestrians, bus and freight operators who cross aging bridges and flyovers every day.]
- [In urban areas, manual vibration sensors... rely solely on experts in the workstation for verification. In rural areas, there are no sensors, checked once or twice a year manually, which leaves intermediate period unchecked, leading to sudden and fatal collapses, like Gambhira disaster in gujarat.]
- [Commuters and public: A daily commuter has no way to know if the bridge is safe. At Morbi, the bridge reopened four days after repairs and about 140 people died. thegoan Nearby residents: A village that depends on one bridge for school, work and hospital visits is suddenly cut off when it collapses or is closed. Gambhira connected two districts, and more than 20 people died when it fell. newsarenaindia Local economy: When a key bridge fails, farmers, shops and transporters lose access to markets. Goods take long detours, costs rise, and rebuilding can take months or years.]

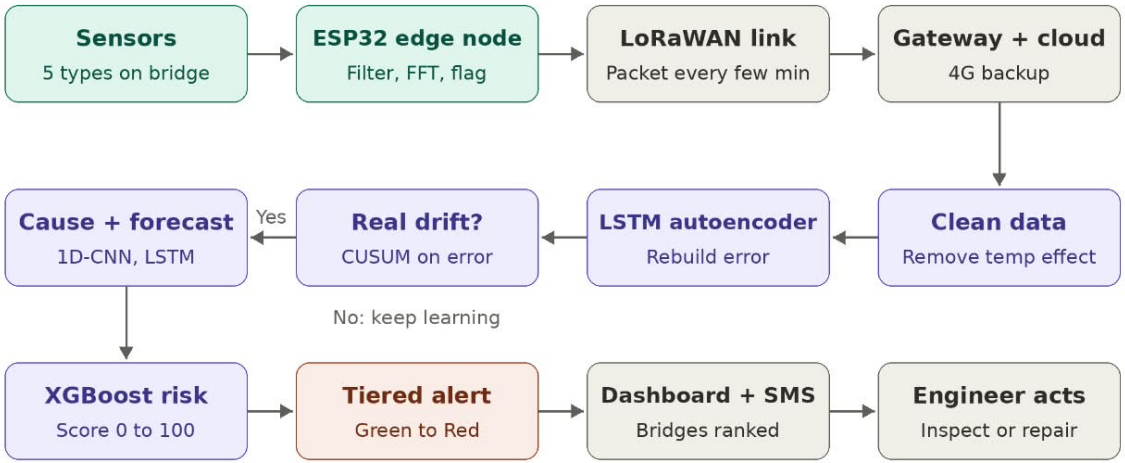
Existing Gaps & Challenges

- [Gap 1: Monitoring covers few bridges. Sensor systems are mandated mainly for national highway bridges, and most bridges worldwide still have no continuous monitoring. Older state and local bridges are left to occasional manual checks.]
- [Gap 2: Reports and data don't force action. Citizen apps like Gujarat's Guj-MARG collect complaints and sensors collect readings. Neither automatically restricts a bridge, and "resolved" is not independently verified.]
- [Main unsolved challenge: Nothing makes a bridge prove it is safe again and again. When evidence is missing or warnings are ignored, it stays fully open by default. (This was a major cause for the gujarat bridge collapse killing 140 people.]

Our Proposed Solution

- [Main idea or product:* A cheap, solar-powered IoT sensor system which constantly monitors ageing bridges and provides an early warning of any structural damage. Each bridge is fitted with sensors that measure vibration, tilt, cracks, and river level, along with an LSTM Autoencoder which learns what normal behaviour is for that particular bridge]
- [In broad terms, the process is as follows: the sensors located on the bridge transmit their data to an ESP32 node, which in turn sends brief data over LoRaWAN to the cloud; the LSTM Autoencoder detects persistent deviations from normal behaviour and does so while disregarding the usual effects of temperature and traffic; the XGBoost model then combines this information with the age of the bridge, traffic data and flood history to produce a risk score, after which the system issues a tiered alert (Green, Yellow, Orange, Red).]
- [The main benefit to users is that engineers and authorities obtain weeks of early warning rather than being limited to once-every-two-years visual inspections, and are given a ranked list of the most risky bridges so that their limited repair funds can be directed where they are most needed. The

Technical Approach



[The above flowchart depicts the working of our safety system]

[Sensors on the bridge (vibration, tilt, cracks, temperature, water level) feed an ESP32 node that cleans the data and sends it over LoRaWAN to the cloud. There, an AI model learns the bridge's healthy behaviour and flags lasting changes while ignoring temperature and traffic effects. Further models identify the likely damage type and forecast its growth, then combine these with age, traffic and flood history into a risk score. Engineers get a Green/Yellow/Orange/Red alert and a dashboard of all bridges ranked by risk.]

Tech Stack

Innovation & Uniqueness

- [*New:* a low-cost, self-learning system for ordinary bridges. Existing monitoring covers only landmark bridges.]
- [*Why it suits the problem:* damage data barely exists, so an unsupervised LSTM Autoencoder learns each bridge's healthy behavior. LoRaWAN and solar power work in remote areas.?]
- [*Key differentiator:* it is affordable and scalable, compensates for temperature and traffic to avoid false alarms, and ranks risk across the whole network.]

Feasibility & Scalability

- **Feasibility:** [Off-the-shelf sensors, ESP32 and LoRa modules. Steps: prototype on a model bridge or the Z24 dataset, run a 2-4 week healthy baseline, pilot on one bridge, then deploy.]
- **Scalability:** [Add nodes per bridge. One gateway covers several km, each bridge trains its own model, and the cloud scales with the network.]
- **Risks:** [Low LoRa bandwidth: process on the edge and send compact packets. False alarms: environment compensation plus CUSUM. No failure data: FEM simulation and public datasets. No coverage: 4G/NB-IoT fallback.]

Cost for Product (Optional)

- Development or hardware cost: [About 15,000-25,000 (sensors, ESP32, LoRa, solar and battery).]
- Hosting or maintenance cost: [About 3,000 per month for the whole network, plus a shared gateway of about 20,000 and battery replacement.]
- Estimated total: [About 25,000 per bridge in year only]

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: [State engineer, pre-monsoon:* continuous tilt and water-level data warns of scour on an old river bridge.]
- Use case 2: [*Highway authority:* the risk ranking shows which of hundreds of bridges to repair first with limited budget.]
- Use case 3: [After a flood, earthquake or overload:* an instant alert tells authorities whether to restrict traffic before anyone is hurt.]
  Note: Additional slides may be included only for this section.

Impacts and Benefits

- [*Users:* engineers get continuous monitoring and early warning instead of twice-a-year visual checks.]
- [*Impact:* saves lives, avoids collapse costs and traffic disruption, and reduces unnecessary rebuilding.]
- [*Measurable targets (to validate):* over 90 percent recall, under 1 false alarm per bridge per month, and weeks of warning lead time.]

Sustainable Development Goals (SDGs)

9	9.1	Industry, innovation and Infrastructure	Resilient reliable, infrastructure through smart monitoring
SDG [11]	Target [11.5]	[Sustainable cities and communities]	[Fewer deaths and losses from structural failure]
SDG [13]	Target [13.1]	[Climate Action]	[Resilience to flood hazards]

Individual Member Contribution

Member	Contribution or responsibility
Mucharla Krishna Chaitanya	[Leader, made and uploaded all ppts,documents, presenter.]
[Sadhu Abhinav]	[Chart preparation, Research papers]
[Achyutha Venkata Manoj]	[Chart preparation, extensive research and solutions.]
Abhiram Rayabaram]	[AI research, supported all activtities]

Yukti ID

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